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ARTICLE



Association between socioeconomic status and prognosis after ischemic stroke in South China

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ABSTRACT

Background: Previous studies have conflicting results on the association between socioeconomic status and stroke outcomes. Thus, this study aimed to investigate whether socioeconomic status is associated with the prognosis of ischemic stroke in Guangzhou, South China.

Methods: A total of 622 patients with ischemic stroke who were admitted in the neurology department of five general hospitals in Guangzhou from May 2014 to October 2014 were included in the study. Socioeconomic status was measured based on education, income, caregiver, and insurance. The modified Rankin scale was used to evaluate the two years prognosis of patients with stroke. A multivariate logistic regression model was performed to determine the association between socioeconomic status and ischemic stroke prognosis.

Results: In 2 years follow-up, 542 ischemic stroke patients were interviewed by telephone. Age ranged from 31 to 96 years, 65.5% of the patients were male. 33.9% of patients with ischemic stroke had a poor prognosis. After adjustment for age, sex, cardiovascular disease, behavior lifestyle, and severity of stroke, odds ratio for poor prognosis in patients with low income was 1.84 (95%CI 1.05–3.22), family caregiver 3.19 (95%CI 1.05–9.70), and no insurance 1.68 (95%CI 1.02–2.77).

Conclusions: Patients with low income, family caregiver, and no insurance have a poorer prognosis after ischemic stroke that can be partly explained by intermediate variable in the patients' demographic characteristics, cardiovascular disease, behavior lifestyle, and stroke severity.

Abbreviations: DALYs: disability-adjusted life years; SES: socioeconomic status; FMC: free medical care; MIUR: medical insurance for urban residents; MIUE: medical insurance for urban employees; NCMS: new rural cooperative medical scheme; NIHSS: National Institute of Health Stroke Scale; mRS: modified Rankin scale; OR: odds ratios; CI: confidence interval

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Introduction

Globally, approximately 25.7 million stroke survivors, 6.5 million deaths from stroke, and 113 million disability-adjusted life years (DALYs) due to stroke, particularly because of ischemic stroke, were recorded [1]. Medication, care, and rehabilitation after stroke do not only cause a financial burden in stroke survivors, family members, and the society but also affect the quality of life of patients and their caregivers [2–5].

Previous studies suggested that socioeconomic status (SES) is also an important potential modifiable risk factor of stroke outcomes [6,7]. Low SES was associated with the increased risk of developing poor prognosis after stroke [8,9]. Numerous population-based studies showed that low education level [10–12], income [6,11,13–15], and occupation [6,11,14,15] may result in the increased risk of developing poor prognosis after stroke. Nonetheless, several studies suggested that there was no relationship between education [14,16], income [10,11,17], occupation [10,11] and stroke outcomes. In addition, the association

between SES and stroke outcomes was not verified due to the differences in country or region and SES indicators.

China is the largest developing country in the world, the incidence rate of ischemic stroke increased by 8.7% annually on average [18]. Stroke is the leading cause of death and disability as defined by DALYs in China [19]. To date, most studies have mainly focused on the pathophysiological factors of stroke and its therapeutic approaches, and the relationship between SES and stroke outcomes in China has not been elucidated. Thus, this study aimed to investigate whether SES is significantly associated with the prognosis of acute ischemic stroke in terms of social and economic factors in individuals in Guangzhou, South China, to improve the prognosis and quality of life of individuals after stroke.

Materials and methods

Study population

This hospital-based cohort study was conducted in South China, and data on patients with ischemic

stroke who were admitted in the neurology department of five general hospitals in Guangzhou from May 2014 to October 2014 were obtained.

Inclusion and exclusion criteria

The inclusion criteria were as follows: (1) patients with ischemic stroke aged ≥ 18 years; (2) patients who visited the hospital within 7 days after the onset of symptoms; (3) patients with ischemic stroke who were diagnosed based on the World Health Organization criteria [20], and those who underwent brain-computerized tomography or magnetic resonance imaging. The exclusion criteria were as follows: (1) patients with impaired consciousness; (2) patients with central nervous system diseases, such as brain tumor, intracranial hemorrhage, subarachnoid hemorrhage, and spinal cord disease.

Ses

Data on education, income, caregiver, and insurance were considered as indicators of SES. The participants were classified into four groups based on the level of education: (1) primary education or below, (2) junior high school, (3) senior high school, (4) university or college education or higher. Individual monthly income was stratified into $< \$301$ and $\leq \$301$. Caregivers consist of formal and informal caregivers [21]. In this study, caregivers based on current national conditions were categorized into family caregiver, health-care assistant, and no caregiver. Insurance according to medical insurance system in China was categorized as: (1) no insurance; (2) free medical care (FMC), a social security system in which the state provides free medical treatment and preventive services to the persons who enjoy them through the medical and health departments; (3) medical insurance including medical insurance for urban residents (MIUR), and medical insurance for urban employees (MIUE), new rural cooperative medical scheme (NCMS) [22].

Other risk factors

Individuals with cardiovascular disease, including hypertension, coronary heart disease, hyperlipidemia, atrial fibrillation, diabetes, and history of stroke and those who were taking medications were diagnosed by professional neurologists. Risk factors of behavior lifestyle were identified such as smoking and alcohol consumption. Smoking is defined as the consumption of more than one cigarette a day and for more than half a year. Alcohol consumption is defined as the consumption of more than 100 g of alcohol per day or more than 500 g a week. The National Institute of Health Stroke Scale (NIHSS) was used to identify stroke severity [23].

Stroke prognosis

The modified Rankin scale (mRS) was used to assess the functional outcome of patients [24]. The mRS is composed of 6 grades, from 0 to 5, with 0 corresponding to no symptoms and 5 corresponding to severe disability. Individuals with an mRS score of 0–2 had a good prognosis. Those with a score of 3–5 had a disability after stroke. Moreover, the newly added mRS score of 6 corresponded to death after stroke. Eventually, participants with an mRS score of 3–6 had a poor prognosis. For each case, the starting point of follow-up was the onset of acute ischemic stroke. The end-point events were functional outcome, death and lost to follow-up within two years after acute ischemic stroke.

Data collection

The questionnaire survey has previously been published elsewhere, and an internal consistency test with a better reliability was carried out using a Cronbach's alpha coefficient of 0.843 [25]. When reliable answers could not be directly obtained from the participants (due to cognitive impairment, communication difficulties, or severe illness), the information was acquired from a family member, caregiver, or hospital staff. The interviews were conducted by clinical nurses with supervision from medical doctors. Training was performed to all the interviewers and supervisors prior to the survey and following-up with telephone interview. If patients refused follow-up or did not answer 3 phone calls within a day for three consecutive days, they were considered lost to follow-up. Eighty out of 622 patients were not contacted 2 years after stroke onset, and the lost to follow-up rate was 12.9%. A verbal informed consent was obtained from the patients or their family members before a face-to-face interview with the participants. Ethical approval for the study was obtained from the ethics committee of First Affiliated Hospital of Guangdong Pharmaceutical University.

Statistical analysis

In this study, to examine the statistical difference for the baseline information between the prognosis groups, a one-way analysis of variance for continuous variables and the chi-square test for categorical variables were carried out. Correlation analysis was conducted to understand the correlation between SES indicator. Multivariate logistic regression models were performed to evaluate the association between SES and stroke prognosis using the adjusted odds ratios (OR) and 95% confidence interval (95% CI). A binomial regression model where the outcomes are good (mRS 0–2) and poor (mRS 3–6) was used in multivariate analysis. Four adjusted models were

analyzed: Model 1 adjusted for age and sex. Model 2 further adjusted for hypertension, coronary heart disease, hyperlipidemia, atrial fibrillation, diabetes, and stroke history. Model 3 further adjusted for smoking, and alcohol consumption. Model 4 further adjusted for stroke severity with NIHSS score. A p-value of 0.05 was considered statistically significant, and all statistical analyses were performed with SPSS 13.0 software package (SPSS, Inc., Chicago, IL).

Results

From May 2014 to October 2016, a total of 542 patients with ischemic stroke were followed up (Figure 1), whose average age was 63.2 years (range: 31–96 years), of which 355 (65.5%) were male. During the 2-year follow-up, 184 (33.9%) patients with ischemic stroke had a poor prognosis. The distribution of all functional outcomes at 2 years for the entire cohort as follows: 0 scores, 1 scores, 2 scores, 3 scores, 4 scores, 5 scores, 6 scores were 192 (35.1%), 113 (20.8%), 55 (10.1%), 63 (11.6%), 39 (7.2%), 10 (1.8%), and 72 (13.3%), respectively.

Differences on the baseline characteristics of the prognosis of ischemic stroke

Table 1 shows the baseline information on demographic factors, SES, cardiovascular disease, behavior lifestyle, and the stroke severity for those with good and poor prognosis after 2 years. There was a significant difference between the two groups on age, education, income, caregiver, hypertension, coronary heart disease, hyperlipidemia, diabetes, atrial fibrillation, stroke history, and NIHSS ($P < 0.05$). Income was significantly correlated with education, insurance ($r = 0.400$, $r = 0.139$, respectively). Correlation was significant at $P < 0.001$.

Table 1. Baseline characteristics of patients according to stroke outcomes.

Variables	Good (n= 358) n/%	Poor (n= 184) n/%	P
Demographic factor			
Sex (male)	237/66.2	118/64.1	0.631
Age (year) mean $\pm$ SD	59.9 $\pm$ 11.9	69.9 $\pm$ 11.5	<0.001
Socioeconomic status			
Education			
Primary education or below	101/28.2	79/42.9	<0.001
Junior high school	147/41.1	76/41.3	
Senior high school	82/22.9	22/12.0	
University/college or higher	28/7.8	7/3.8	
Income (\$)			
<301	42/11.7	44/23.9	<0.001
301–603	316/88.3	140/76.1	
Caregiver			
Family caregiver	305/85.2	169/91.8	0.005
Health care assistant	15/4.2	10/5.5	
No caregiver	38/10.6	5/2.7	
Insurance			
No insurance	89/24.9	53/28.8	0.243
Free medical care	19/5.3	7/3.8	
Medical insurance	250/69.8	124/67.4	
Vascular risk factor			
Hypertension	195/54.5	122/66.3	0.008
Coronary heart disease	15/4.2	25/13.6	<0.001
Hyperlipidemia	48/13.4	15/8.2	0.045
Atrial fibrillation	5/1.4	13/7.1	<0.001
Diabetes	75/20.9	60/32.6	0.003
Stroke history	52/14.5	56/30.4	<0.001
Behavior lifestyle			
Smoking	116/32.4	50/27.2	0.212
Alcohol consumption	45/12.6	26/14.1	0.610
Stroke severity			
NIHSS (score) mean $\pm$ SD	3.3 $\pm$ 4.0	6.7 $\pm$ 5.7	<0.001

Good: mRS (0–2); Poor: mRS (3–6).

NIHSS: National Institute of Health Stroke Scale.

Relative risk of poor prognosis with ischemic stroke in terms of SES

In Table 2, we found that family caregiver, and no insurance coverage were associated with poor prognosis of ischemic stroke in Model 1, adjusted OR was 3.43 (1.28–9.19), 1.65 (1.06–2.58), respectively. In Model 2, further adjusted for cardiovascular disease, patients with low income, family caregiver, and no insurance had a poor prognosis, adjusted OR was

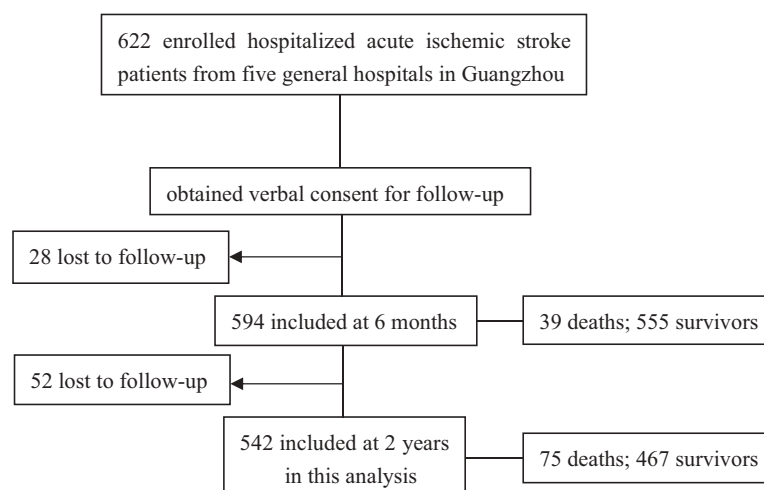


Figure 1. Patients flow chart.

Table 2. Odds ratio for the socioeconomic status of individuals with stroke.

	Model 1	Model 2	Model 3	Model 4
Education				
Primary school or below	1.78(0.69–4.61)	1.99(0.73–5.41)	2.10(0.78–5.67)	2.22(0.74–6.62)
Junior high school	2.04(0.79–5.23)	1.98(0.74–5.32)	2.11(0.79–5.63)	2.43(0.83–7.17)
Senior high school	1.69(0.60–4.76)	1.86(0.63–5.48)	1.98(0.68–5.81)	2.52(0.78–8.14)
College or higher	1.0	1.0	1.0	1.0
Income (\$)				
<301	1.54(0.93–2.54)	1.74(1.04–2.92)*	1.77(1.05–2.98)*	1.84(1.05–3.22)*
≥301	1.0	1.0	1.0	1.0
Caregiver				
Family caregiver	3.43(1.28–9.19)*	3.09(1.14–8.39)*	3.21(1.18–8.72)*	3.19(1.05–9.70)*
Health care assistant	2.58(0.70–9.52)	2.15(0.56–8.27)	2.25(0.58–8.67)	1.16(0.24–5.71)
No caregiver	1.0	1.0	1.0	1.0
Insurance				
No insurance	1.65(1.06–2.58)*	1.64(1.03–2.61)*	1.61(1.01–2.57)*	1.68(1.02–2.77)*
Free medical care	0.88(0.59–1.33)	0.54(0.19–1.52)	0.53(0.19–1.48)	0.61(0.20–1.81)
Medical insurance	1.0	1.0	1.0	1.0

Model 1: adjusted for sex and sex. Model 2: further adjusted for hypertension, coronary heart disease, hyperlipidemia, atrial fibrillation, diabetes, stroke history. Model 3: further adjusted for smoking, alcohol consumption. Model 4: further adjusted for NIHSS. *: $P < 0.05$.

1.74 (1.04–2.92), 3.09 (1.14–8.39), 1.64 (1.03–2.61), respectively. In Model 3, further adjusted for smoking and alcohol consumption, patients with low income, family caregiver, and no insurance had a poor prognosis, adjusted OR was 1.77 (1.05–2.98), 3.21 (1.18–8.72), 1.61 (1.01–2.57), respectively. In Model 4, further adjusted for stroke severity, risk of poor prognosis in the lowest income group, family caregiver group, and no insurance coverage group increased gradually, adjusted OR was 1.84 (1.05–3.22), 3.19 (1.05–9.70), 1.68 (1.02–2.77), respectively. However, we failed to find that the education was no associated with risk of poor prognosis of ischemic stroke.

Discussion

The effect of SES on stroke outcomes plays a remarkable role in individuals around the world [9]. The present study found that the low income, family caregiver, and no insurance were significantly associated with risk of poor prognosis of ischemic stroke. However, association between low education and poor prognosis of ischemic stroke was failed to find, although the education correlated with income. The risk factors including age, sex, history of disease, smoking, alcohol consumption, and NIHSS can partly explain the association between SES and poor prognosis of ischemic stroke.

The association of low income with prognosis in people with ischemic stroke was observed in our study, but low education not found in Guangzhou, South China. A hospital-based cohort study in China showed that those patients in low education, low income, and socioeconomic deprivation group had a higher one-year mortality after ischemic stroke [11]. Singh et al. performed a population-based study in the United States also found that low education and income can increase mortality of ischemic stroke during the two-year follow-up [15]. Belleudi

et al. also found that mortality of ischemic stroke with low education was higher [26]. However, several studies reported that low income was not associated with an increased mortality of ischemic stroke, and association of low education was not observed [14]. This difference probably because of the complexity of the SES indicators, the differences in the area or population, and the population simple size.

Results of the study showed that a family caregiver was associated with the increased risk of developing a poor prognosis compare with no caregivers, but association between health-care assistant and prognosis was not observed. Patients without caregivers had a slight neurological functional impairment, with a stronger self-care ability in daily life. Therefore, those patients had a good prognosis. The health-care assistant is employed by patients or their relatives to provide daily care or for patients. However, only a small percentage of the population hires health-care assistant to care for patients, depending on their socioeconomic status. If the family is in good socioeconomic condition, they would have the ability to afford additional care costs. On the contrary, patients have to choose their family caregiver to care for them. Family caregiver (relatives or friends) is informal caregiver, which have nature of society and economics [27]. Family culture is very important for Chinese, and a family caregiver is a representative of that culture [28]. Most of the patients with stroke suffer from varying degrees of functional impairment after discharge. Family caregivers take a large amount of responsibilities for stroke survivors at home. A US-based study on the family caregivers of patients with multiple chronic conditions, disability, and cognitive impairment, caregivers play an important role [29]. During long-term care, the presence of psychological distress, depression of family caregivers impacts the quality of care, also indirectly affect the poor prognosis of stroke [30,31].

In our study, patients with no insurance was associated with poor prognosis of ischemic stroke.

Insurances reflect the distribution of financial resources and the degree of social support from the country, and it also reflect education, income, and occupation of individual. In China, the medical insurance derives from government-sponsored social health insurance schemes [22]. A study reported that uninsured patients had a higher level of neurological impairment and higher risk of mortality, and the mortality of ischemic stroke was higher than that of intracerebral hemorrhage [32]. Patients who have insurance are more likely to have access to health service, resulting to positive outcomes, and uninsured patients are less likely to receive health benefits, contributing to poor outcomes. A study in the United States reported that uninsured patients were less likely to receive preventative care prior to ischemic stroke, time to presentation for acute treatment, and access to rehabilitation, contributing a higher mortality of ischemic stroke [33]. Thus, further studies should be investigated in terms of insurance.

Some hypothesis can partly explain the association between SES and prognosis of ischemic stroke, in accordance with this study. First, low SES have more poor behavior lifestyle such as excessive smoking, unhealthy diet, alcohol consumption, lack of physical activities, resulting in poor prognosis [11,34]. Second, psychological distress, depression, negative emotions can mediate the association between SES and prognosis [35,36]. Third, hypertension, diabetes, hyperlipidemia, cardiovascular diseases were risk factors of ischemic stroke, and controlling those risk factors can improve prognosis of ischemic stroke [5]. Therefore, vascular risk factors are likely to play an important role in association between SES and stroke mortality. Fourth, low SES failed to receive good health service, which can increase risk of poor prognosis [37,38]. For example, those patients with low SES before onset of ischemic stroke lack of awareness of primary prevention. So, they less or never check Blood pressure, blood lipid, blood glucose, and others. Besides, once ischemic stroke occurs, patients with stroke lack of awareness of timely seeking to hospital and delayed the best treatment time. In the treatment process, patients with low education fail to communicate effectively with doctors or nurses and have a poorer medication compliance such as the presence of drug withdrawal or reducing. Therefore, we should further investigate the mechanism of association between SES and ischemic stroke outcomes in the future.

Limitations

This study had several limitations. First, the sample was limited by hospital-based population, a larger, multicentric, perspective cohort study is further warranted. Using structure questionnaires, it is difficult to understand the intention, motivation and thinking processor of the patients. We try to establish trust with patients before interview, so that patients can answer the

questionnaire more truthfully. The follow-up data were collected by telephone. It can be easy to generate lost to follow-up. Besides, interviewer cannot objectively assess patient functional outcome. Second, occupation was not included in the analysis. This is primarily due to the complexity of indicator, lack of a classification standard, and most of the patients were retirees in China, contributing inaccurate data. Third, the importance of thrombolytic therapy for the risk of developing a poor prognosis was not considered. Based on our knowledge, a timely thrombolytic therapy can reduce functional impairment after ischemic stroke. Moreover, we failed to investigate the impact of psychological factors and exposure environment on association between SES and prognosis of ischemic stroke.

Conclusions

The study demonstrated that low income, family caregiver, and no insurance were significantly associated with poor prognosis of ischemic stroke 2 years after stroke in Guangzhou, South China. Our findings also support conclusions from previous studies indicating socioeconomic differences in stroke outcomes could be mainly mediated by differences in demographic characteristics, vascular risk factors, behavior lifestyle, stroke severity. Concerning psychological factors, provision of health care need to further explain and adopt interventions to reduce these inequalities.

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Author Contributions

WSP, WMT, and WJ designed the study and generated the questionnaires. WSP, WMT, and SBY conducted data collection and data analyses. WL and WJ provided critical review of the manuscript. All authors participated in the interpretation of the data and approved the final draft.

Statement of Ethics

Subjects (or their parents or guardians) have given their written informed consent.

The study protocol has been approved by ethics committee of First Affiliated Hospital of Guangdong Pharmaceutical University.

Disclosure statement

No potential conflict of interest was reported by the authors.

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